

Projection (π) Operator in Relational Algebra

EXAM-READY STUDY NOTES & REVISION GUIDE

1 What is Projection?

Projection is a fundamental unary operator in Relational Algebra represented by the Greek symbol π (**Pi**).

★ EXAM-READY DEFINITION

Projection is a unary relational algebra operation represented by π that retrieves/selects specified attributes (columns) from a relation and automatically eliminates duplicate tuples.

Syntax:

$$\pi_{\text{attribute1, attribute2, ...}}(\text{Relation})$$

Basic Example:

$$\pi_{\text{Roll_No}}(\text{Student})$$

Meaning: Retrieve only the **Roll_No** column from the **Student** relation.

2 Example of Single Column Projection

Consider the initial **Student** table:

Roll_No	Name	Age
1	A	20
2	B	21
3	A	20

Query: $\pi_{\text{Roll_No}}(\text{Student})$

Roll_No
1
2
3

3 Projection of Multiple Columns

We can specify more than one column in the subscript attribute list:

Query: $\pi_{Roll_No, Name}(Student)$

Roll_No	Name
1	A
2	B
3	A

- π_{Roll_No} → Retains 1 column
- $\pi_{Roll_No, Name}$ → Retains 2 columns
- $\pi_{Roll_No, Name, Age}$ → Retains all 3 columns

4 Most Important Property: Duplicate Removal ★★★★★

Relational algebra is based on **mathematical set theory**. Relations are sets of tuples, and sets cannot contain duplicate elements. Therefore, **Projection automatically eliminates duplicate tuples**.

Consider the **Student** table again:

Roll_No	Name	Age
1	A	20
2	B	21
3	A	20

Query: $\pi_{Name}(Student)$

Raw attribute values extracted: A, B, A. Since A is repeated, set semantics retain it only once.

Name
A
B

🏆 **Golden Rule: Projection (π) = Select Columns + Remove Duplicate Tuples**

5 Multi-Column Duplicate Removal Nuance

Consider evaluating: $\pi_{Roll_No, Name}(Student)$

The output tuples are: $(1, A)$, $(2, B)$, $(3, A)$.

Although 'A' appears twice in the `Name` attribute, the *complete projected tuples* $(1, A)$ and $(3, A)$ are distinct because their `Roll_No` values differ.

⚠ EXAM TRAP WARNING

Do **not** remove a row simply because one individual column contains repeated values! Duplicates are eliminated based strictly on the **entire projected tuple**.

6 Non-Destructive Nature of Projection

Executing $\pi_{Name}(Student)$ generates a **new temporary result relation**. It does *not* modify, delete, or alter the original `Student` table in the database.

7 Projection (π) vs. Selection (σ)

This is one of the most frequently asked comparisons in DBMS exams:

Operator	Symbol	Target Focus	Core Function
Selection	σ (Sigma)	Rows (Tuples)	Filters horizontal rows based on a specified predicate condition.
Projection	π (Pi)	Columns (Attributes)	Selects vertical columns and eliminates duplicate rows.

🧠 **Memory Trick:** Selection = Subset of Rows | Projection = Pick Columns

Code / Expression Comparison:

- $\sigma_{Age > 20}(Student)$ → Filters rows where `Age > 20`.
- $\pi_{Name, Age}(Student)$ → Selects columns `Name` and `Age`.

8 Projection in Complex Expressions

In complex queries, Projection is commonly applied as the outer/final wrapper to dictate which attributes appear in the final output.

$$\pi_{Name}(\sigma_{Age > 20}(Student))$$

Evaluation Sequence:

Filter Rows ($\sigma_{Age > 20}$) → Select Columns (π_{Name}) → Final Result

9 Mathematical Formalism

If relation R has schema $R(A, B, C, D)$, then the projection:

$$R' = \pi_{A, C}(R)$$

produces a result relation $R'(A, C)$ containing only attributes A and C with all duplicate tuples eliminated under mathematical set semantics.

✓ Key Exam Takeaways & Summary

- **Symbol:** Projection is denoted by π (Pi).
- **Dimension:** Operates strictly on **attributes/columns**.
- **Set Semantics:** Automatically eliminates **duplicate tuples**.
- **Original Data:** Non-destructive; original table remains unchanged.
- **Tuple-Level Uniqueness:** Duplicates are evaluated across the *complete combination* of projected attributes.

🚀 Quick Revision Chart

RELATIONAL ALGEBRA
↓
PROJECTION (π)
↓
Pick Specific Columns
↓
Eliminate Duplicate Tuples

One-Line Summary: Projection (π) retrieves specified attributes (columns) from a relation while removing all duplicate tuples.